Install • Run • Stop — Java Servers

Tomcat 10.1 • GlassFish 7.0 • WildFly 30 — Jakarta EE 10 • Dark Modern v3

Lab: ~/Documents/AppServers/ | Java 17 + Maven | /opt/tomcat /opt/glassfish7 /opt/wildfly
Rule: port 8080 shared → run only ONE server at a time.

0 — Build WAR (all)

```
cd ~/Documents/AppServers/tomcat/jakarta-10
mvn package
cd ~/Documents/AppServers/glassfish/jakarta-10
mvn package
cd ~/Documents/AppServers/wildfly/jakarta-10
mvn package
```

1 — Tomcat · The Scooter

Lightweight Servlet container. /opt/tomcat 10.1.60

▸ Run

```
cd ~/Documents/AppServers/tomcat/jakarta-10 && mvn package
sudo cp target/*.war /opt/tomcat/webapps/
sudo chown -R tomcat:tomcat /opt/tomcat/webapps
sudo /opt/tomcat/bin/startup.sh
# http://localhost:8080/tomcat-jakarta-10-1.0/hello
```

▸ Stop

```
sudo /opt/tomcat/bin/shutdown.sh
ss -tlnp | grep 8080
tail -f /opt/tomcat/logs/catalina.out
```

2 — GlassFish · Swiss Army Knife [FIXED]

Full Jakarta EE 10. Fix permission denied: chown + chmod below.

▸ Install — corrected

```
java -version  # need 17+
sudo pacman -S --needed unzip curl
cd /tmp
curl -L -o glassfish.zip https://github.com/eclipse-ee4j/glassfish/releases/download/7.0.15/glassfish-7.0.15.zip
sudo unzip -q -o glassfish.zip -d /opt/
sudo chown -R $USER:$USER /opt/glassfish7
chmod +x /opt/glassfish7/bin/asadmin
/opt/glassfish7/bin/asadmin version
```

▸ Run

```
sudo /opt/tomcat/bin/shutdown.sh  # free 8080
/opt/glassfish7/bin/asadmin start-domain domain1
cd ~/Documents/AppServers/glassfish/jakarta-10 && mvn package
/opt/glassfish7/bin/asadmin deploy --force --name glassfish-jakarta-10 target/*.war
# http://localhost:8080/glassfish-jakarta-10-1.0/hello
```

▸ Stop

```
/opt/glassfish7/bin/asadmin undeploy glassfish-jakarta-10
/opt/glassfish7/bin/asadmin stop-domain domain1
```

3 — WildFly · Never Sleeps

Jakarta EE + clustering. /opt/wildfly 30.0.1.Final

▸ Install

```
cd /tmp
curl -L -o wildfly.zip https://github.com/wildfly/wildfly/releases/download/30.0.1.Final/wildfly-30.0.1.Final.zip
sudo unzip -q -o wildfly.zip -d /opt/
sudo chown -R $USER:$USER /opt/wildfly-30.0.1.Final
sudo ln -sfn /opt/wildfly-30.0.1.Final /opt/wildfly
```

▸ Run

```
sudo /opt/tomcat/bin/shutdown.sh
/opt/glassfish7/bin/asadmin stop-domain domain1
cd ~/Documents/AppServers/wildfly/jakarta-10 && mvn package
/opt/wildfly/bin/standalone.sh &
cp target/*.war /opt/wildfly/standalone/deployments/
```

▸ Stop

```
/opt/wildfly/bin/jboss-cli.sh --connect :shutdown
rm -f /opt/wildfly/standalone/deployments/wildfly-jakarta-10*.war*
```

Appendix — How to Force Kill App (all servers)

Use when stop fails, port 8080 stuck, or process hung. Order: graceful -> kill -> kill -9 -> free port.

▸ A. Check what holds the port

```
ss -tlnp | grep -E '8080|4848|9990'
ps aux | grep -E 'catalina|glassfish|wildfly|jboss' | grep -v grep
jps -l  # java PIDs
```

▸ B. Tomcat force kill

```
sudo /opt/tomcat/bin/shutdown.sh
sudo systemctl stop tomcat
ps aux | grep catalina | grep -v grep
sudo kill <PID>
sudo kill -9 <PID>  # force
pgrep -af catalina  # verify name first
sudo pkill -9 -f Bootstrap  # Tomcat Bootstrap
sudo fuser -k 8080/tcp  # free port
```

▸ C. GlassFish force kill

```
/opt/glassfish7/bin/asadmin stop-domain domain1
/opt/glassfish7/bin/asadmin list-domains
ps aux | grep glassfish.jar | grep -v grep
kill <PID>
kill -9 <PID>  # force
# avoid: pkill -f glassfish.jar (may match own shell) — use:
pgrep -af glassfish.jar
kill -9 $(pgrep -f 'ASMain.*domain1')
sudo fuser -k 8080/tcp
sudo fuser -k 4848/tcp
```

▸ D. WildFly force kill

```
/opt/wildfly/bin/jboss-cli.sh --connect :shutdown
ps aux | grep -i wildfly | grep -v grep
kill <PID>
kill -9 <PID>  # force
pgrep -af standalone.sh
pkill -9 -f 'jboss.modules'
sudo fuser -k 8080/tcp
sudo fuser -k 9990/tcp
```

▸ E. Verify clean

```
ss -tlnp | grep -E '8080|4848|9990'  # must be empty
ps aux | grep -E 'catalina|glassfish|wildfly' | grep -v grep
```

```
tail -n 20 /opt/glassfish7/glassfish/domains/domain1/logs/server.log
tail -n 20 /opt/tomcat/logs/catalina.out
```